



DSS 325 Open Source Programming Lang. Course Syllabus [‡]

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Lectures:	TR: 8-9:15 am MV209

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COURSE DESCRIPTION

This course introduces students to the fundamentals of Python programming, equipping them with problem-solving skills and the ability to write efficient, readable code. Using Python, students will explore key concepts in programming, including variables, control flow, functions, data structures, and object-oriented programming. The course will also include applications to data analysis and visualization, preparing students for advanced topics or real-world use cases.

Topics covered include:

- Introduction to Python Programming
- Variables and Basic Data Types
- Strings and String Operations
- Control Flow (Conditional Statements and Loops)
- Functions and Modularity
- Working with Data Structures (Lists, Dictionaries, and Tuples)
- File Handling and Exceptions
- Basics of Recursion
- Data Science Basics with NumPy and Pandas
- Visualization Techniques in Python
- Real-World Applications in Python

COURSE OBJECTIVES

By the end of this course, students will:

- Develop an understanding of Python syntax and semantics.
- Use Python to solve computational problems.
- Understand and implement control flow mechanisms.
- Write modular code using functions and classes.
- Manipulate and analyze data using Python's built-in and external libraries.
- Perform basic data visualization using Python.
- Apply Python programming to real-world case studies.

COURSE POLICIES

Required Course Materials:

- **Textbook:** *Introduction to Python Programming* by OpenStax (2024 Edition)
 - Access for free at openstax.org
 - ISBN-13: 978-1-961584-45-7
 - You also can download the free textbook from Canvas.
- Python Documentation: <https://docs.python.org/3/>
- Other readings and resources as assigned on Canvas.

Required Software:

- Download Anaconda and Jupyter Notebook is required. The download instruction can be found from Canvas.

EXAMS AND GRADING CRITERIA:

Grading Criteria:

	%
In-Class Quizzes (Every Chapter)	25%
In-Class Practice	5%
Assignment	10%
Mid-term Exam	25%
Final Project	30%
Attendance/Class Participation	5%
Total	100%

Exam content will be announced in class.

Missed exam/quizzes Students are expected to be present for all assignments, examinations as listed on the syllabus or announced by the instructor. Make-up exams require valid documentation (this includes medical and other family emergencies). Valid documentation of an absence must be requested through the Dean's Office and presented to the instructor in order to reschedule an exam. ***There will be no exceptions made to this policy.*** Please do not request makeup exams or extended assignment deadlines based on family or leisure travel because your request ***will not*** be accommodated. Consult the University's Academic Calendar and plan accordingly

Final Grade Distribution:

A	95-100 points	C+	77-<80 points
A-	90 - <95 points	C	73-<77 points
B+	87-<90 points	C-	70-<73 points
B	83-<87 points	D	65-<70 points
B-	80-<83 points	F	0-< 65 points

Note: Points round up to the nearest 0.5. (Eg: 86.4 is a B, 86.6 is a B+) Canvas provides a “dumb”/unweighted average, which you should ignore. I am unable to delete that column from the Canvas Gradebook

Deliverables

All deliverables are to be submitted through Canvas according to the deadlines stated in Canvas and/or communicated in class.

Laptop Use

Students are **required** to bring laptops to every class for appropriate course-related activities. However, laptop and cellphone use in class is **prohibited** outside of the time slots set up specifically by the instructor. Exceptions may be granted by the teacher on a case-by-case basis.

Email Response Time:

During the week, emails will be returned within 24 hours. Any emails sent on weekends will be returned by Sunday evening if not before.

Use of Canvas and University email:

Canvas / SJU email will be the preferred method of communication for class-related announcements, events, changes, etc. It is the student's responsibility to check Canvas and his/her University email account regularly. Instructor is not responsible for using email accounts other than the student's assigned University account. When email instructor, please specify your section.

Students With Disabilities:

Reasonable academic accommodations may be provided to students who submit appropriate documentation of their disability. If students have need of assistance or questions with this issue, they are encouraged to contact the Office of Student Disability Services (SDS) at sds@sju.edu or by phone at 610.660.1774. The Office of SDS also provides an appeal/grievance procedure for complaints regarding requested or offered reasonable accommodations. More information can be found at: www.sju.edu/sds.

Class Recording:

Be advised that by taking this course you consent to some or all class sessions being recorded by the instructor (and in some cases with assistance by the SJU Office of Information Technology). Recordings are intended to focus on the instructor but the voices and/or likenesses of other individuals in the class may also be recorded. Video and/or sound recordings will be stored at a secure SJU video management platform for two years and will be used or distributed only for educational or academic purposes, e.g., to allow for remote learning for those students not physically present in class or unable to attend a synchronous online class session. If you have questions about the use of such technology or you do not consent to being recorded, you must discuss and resolve your concerns with your instructor before the end

of the drop/add period.

Bias, Discrimination, & Harassment:

As a Catholic, Jesuit University, Saint Joseph's is committed to providing a workplace and educational environment, as well as other benefits, programs, and activities, that are free from acts of bias, discrimination, harassment, including acts of sexual harassment, sexual assault, dating and domestic violence, and stalking. We encourage anyone who has experienced this type of harm to seek help from the University by filing a report through the [incident reporting form](#), or by calling 610-660-1145.

For information about the University's response to reports of bias, harassment, and discrimination, including definitions of prohibited conduct, information regarding confidential reporting options on- and off-campus, and resources for support, please visit <https://www.sju.edu/offices/titleix-equity> or contact titleix@sju.edu / bias@sju.edu.

While I want you to feel comfortable coming to me for support, please know that I have some reporting requirements that are part of my job responsibilities at Saint Joseph's University. For example, I am required to report any student disclosures of bias, discrimination, and harassment. I will keep the information as private as I can, but I am required to bring it to the attention of the institution's Director Office of Title IX, Equity & Compliance.

Anonymous Student Feedback

The following link allows students to provide anonymous feedback directly to the Decision & System Sciences Department (DSS). Students who believe the situation they are experiencing warrants immediate attention can notify the Department via this anonymous survey. Students may also email the Department Chair (Dr. Marcello Balduccini) directly at mbalducc@sju.edu. Please note that this link is not meant to be used as a faculty evaluation and does not replace the student evaluation at the end of the semester.

https://sju.co1.qualtrics.com/jfe/form/SV_4NSi1TCKdsHWF5X

Responsible Employee Reporting:

While I want you to feel comfortable coming to me with issues you may be struggling with or concerns you may be having, please be aware that I have some reporting requirements that are part of my job responsibilities at Saint Joseph's University. For example, if you inform me of an issue of sexual harassment, discrimination or sexual misconduct (Sexual Assault, Sexual Harassment, Sexual Exploitation, Domestic Violence, Dating Violence, or Stalking) I will keep the information as private as I can, but I am required to bring it to the attention of the institution's Title IX Coordinator, Mary- Elaine Perry. If you would like to speak with her directly, she can be reached at mperry01@sju.edu or 610-660- 1145.

Counseling and Psychological Services (CAPS) can provide confidential support and can assist you with determining how you would like to address this issue. CAPS is located at 504A Merion Gardens and can be reached at 610-660-1090. They also have walk-in hours M-F 11- 12:30 in LaFarge. REPP, Rape Education and Prevention Program provides a 24 hour confidential helpline (610-733-9650) as well. For more information about your confidential options at SJU, as well as resources in the community go to the support website.

Academic Honesty:

Saint Joseph's University encourages the free and open pursuit of knowledge; we consider this to be a fundamental principle and strength of a democratic people. To this end, SJU expects its students, its faculty, its administrators, and its staff to uphold the highest standards of academic integrity. The University expects all members of the University community to both honor and protect one another's individual and collective rights.

The SJU Academic Honesty Policy, which is published both in the Student Handbook and in the University Catalog, will be strictly enforced. It is your responsibility to understand this policy. Use of unauthorized notes or assistance during an exam will result in at least failure of the assignment, if not failure of the course. Plagiarism will also result in at least failure of the assignment, if not failure of the course. Plagiarism occurs in any instance where one submits as one's own work statements or ideas taken from another source but not properly acknowledged as such. Any material copied or derived from another source must be acknowledged by giving a reference. For direct quotes, always use quotation marks and provide a reference. If you paraphrase something, you must provide a reference to acknowledge where you found the material. Internet sources, if used, must be cited.

Students are expected to include the following Academic Honesty Affirmation on each of their assignments: "I attest to the honesty of my academic work and affirm that it conforms to the standards of the Academic Honesty Policy of Saint Joseph's University."

AI Tools Usage Policy:**Scope**

- **AI Tools Covered:** This includes, but is not limited to, machine learning models, chatbots (e.g., ChatGPT), AI content generators, and automated problem solvers.
- **Applicability:** The policy applies to all assignments, quizzes, exams, and projects unless specified otherwise.

Guidelines

- **Disclosure Requirement:** Students must disclose the use of any AI tools in completing coursework. Submissions must include a statement or footnote indicating the AI tools used and the extent of their involvement.

Prohibited Uses

- Code Generation: **Students are prohibited from using AI tools to write code for assignments, quizzes, or projects.** The intent is to assess students' own understanding and capability to write and debug code.
- Exam Integrity: AI tools are not permitted to assist in answering questions on exams and quizzes.

Permitted Uses

- Debugging Assistance: Students may use AI tools to identify errors in their code and receive suggestions for correction.

Ethical Considerations

- Integrity: Use of AI must adhere to the principles of academic honesty, ensuring submissions are the original work of the student.
- Privacy and Compliance: AI use must not infringe on the rights of others or violate any laws regarding data privacy and intellectual property.

Consequences

- Non-compliance with this policy will be considered a serious violation of academic integrity and will result in disciplinary actions as outlined in the student handbook.

Discussion and Updates

- Students are encouraged to discuss the use of AI tools with the instructor to clarify any uncertainties about appropriate applications.
- This policy will be periodically reviewed and may be updated to reflect new developments in AI technologies and applications.

Changes to Syllabus:

This syllabus is subject to change at the sole discretion of the instructor. Any changes will be announced class- wide in a timely manner and posted on the class website.

Inclusivity Statement:

I am committed to providing an atmosphere for learning that respects diversity. While working together to build this community I ask that we all:

- Recognize that our classroom community represents a rich variety of backgrounds and perspectives.
- Share our unique experiences, values, and beliefs.
- Be open to the views of others.
- Honor the uniqueness of our community.
- Appreciate the opportunity that we have to learn from each other in this community.
- Value each other's opinions and communicate in a respectful manner.
- Keep confidential discussions that the community has of a personal (or professional) nature.
- Use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the SJU community.